Samer Aslan

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ML engineer and technical lead of Bloomberg's first AI agent for law, from prototype to 200,000+ users, and neuro-AI researcher with the Kriegeskorte lab at Columbia.

Experience

Bloomberg LP | Machine Learning Engineer — *Feb 2024 to Present*

Conversational (Agentic) AI, Bloomberg Law AI

Recruited to the team of senior ML engineers after spending nights and weekends building the guardrails that now protect Bloomberg Law AI.

- Conceived and prototyped Bloomberg's first AI agent for law, which carries an attorney through an entire case, won buy-in and led 15+ engineers and attorneys as technical lead. Now Bloomberg's flagship AI product for law, live for 200,000+ users.
- Owned and built the agent's core harness on LangGraph: tools and skills, sub-agent delegation, context assembly and compaction for long-running cases, cross-session memory, observability and scaling.
- Built the industry's first deep research system for tax law, which cut multi-day legal research to minutes for the 20,000+ Bloomberg Tax users.
- Developed Bloomberg Law AI's question answering and deep research agents, which reason over up to 100,000 user-uploaded documents.
- Trained and deployed a cross-encoder re-ranking model for agent search over 2 billion+ documents, reducing cost by 90%+ versus LLMs.
- Engineered Pulse, Bloomberg Law's first automated evaluation platform, which scores outputs with LLM-as-judge and gates releases for all its AI agents. It cut the time to evaluate a change from 2-3 weeks of manual testing to 2 hours.
- Built a neuro-symbolic system to improve LLM legal reasoning, in which an LLM rewrites tax law as code that runs deterministically. It raised tax-case accuracy from 55% to 85% and held up on a new benchmark of altered cases.

Bloomberg Terminal, FX (Currency) Pricing | Software Engineer

- Rebuilt the real-time distributed system that reconciles competing bank data into Bloomberg's authoritative currency rate for 300,000+ Terminal users, and raised its throughput 5x at a 5 ms update cycle.
- Developed a full-stack configuration platform for FX and commodity pricing, used daily by 250+ analysts and product staff.

Bloomberg Terminal, Mortgages | Software Engineering Intern (Summer 2022)

Columbia University | Researcher — *Jul 2025 to Present*

Visual Inference Lab (Kriegeskorte), Zuckerman Institute | Part-time, alongside full-time role

- Developed UniMap, an algorithm for mapping high-dimensional data where every point gets equal space and similar points cluster. Unlike in t-SNE or UMAP, a region's size shows how many points it holds. Now writing the paper.
- Built a new exact, near-linear UniMap core algorithm that cut map generation from minutes to seconds, up to 20x faster end to end.
- Applied UniMap and representational similarity analysis to AI safety and interpretability, mapping where an open LLM's internal refusal signal differs from its actual refusals (sameraslan.com/notes/unimap-interpretability-vqtoip).
- Mapped 5,000 Americans by their full political views with UniMap, one pixel per person, and shipped a full-stack website that runs UniMap in the browser on datasets from Venn diagrams to music genres.
- Analyzed 1,024-electrode recordings from three successive stages of macaque visual cortex (V1, V4, IT) and found that object categories become easier to tell apart in the neural responses at each later stage.

Publications and Talks | scholar.google.com/citations?user=HGDfgIQAAAAJ

- **Paper:** *Reasoners or Translators? Contamination-aware Evaluation and Neuro-Symbolic Robustness on Tax Law*. SURGeLLM Workshop at ACL 2026, co-first author. arXiv 2605.16052.
- **Talk:** Selected from 20,000+ employees as one of three engineers to present to Michael Bloomberg and senior leadership on the safe, reliable and effective use of AI. Showcased Bloomberg Law AI.
- **Paper:** *PercepLog-0: LLM-Augmented Backward Chaining for Legal Reasoning*. Findings of EMNLP 2026 (accepted).
- **Paper:** *DiaLong: Benchmarking Reminiscence in Long-Context Dialogues*. Preprint, 2023, co-first author. arXiv 2610.25463.
- **Talks:** Presented Bloomberg Law AI to the 200+ person Bloomberg Law, Tax and Government engineering group, and the *Reasoners or Translators?* ACL paper to Bloomberg's 1,000+ member ML community.

Selected Projects

- Launched Tansa (tansa.app) to help listeners explore new music beyond their usual taste. It maps 10,000+ acclaimed albums by how they sound and feel rather than listening history or genre tags by converting album audio into embeddings.
- Trained a real-time model that generates sound for silent video of musical scenes, and improved its audio-visual sync with a training loss modeled on the neuroscience of how much lag humans tolerate (github.com/sameraslan/PerceptAlign).

Skills

Languages: Python, C++, TypeScript, JavaScript, SQL, MATLAB
ML and agents: PyTorch, scikit-learn, NumPy, LangGraph, LangChain, LangSmith, rsatoolbox
Infrastructure and web: Docker, SLURM, PostgreSQL, React, Next.js, Node.js

Education

Johns Hopkins University — *Dec 2023*

M.S. Computer Science, GPA 4.0/4.0. B.S. triple major in Computer Science, Cognitive Science, and Computer Engineering.